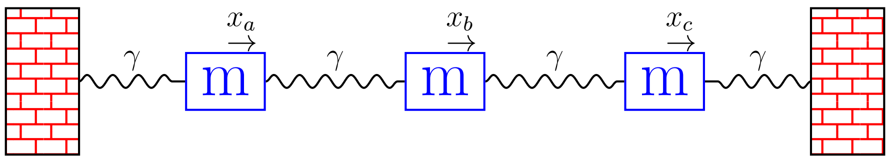
Homework 1

Coupled Oscillators

1. Conider the following free Hamiltonian:

$$H = \frac{p_a^2}{2m} + \frac{p_b^2}{2m} + \frac{p_c^2}{2m} + \frac{1}{2}\gamma(x_a^2 + x_c^2) + \frac{1}{2}\gamma(x_b - x_a)^2 + \frac{1}{2}\gamma(x_c - x_b)^2$$



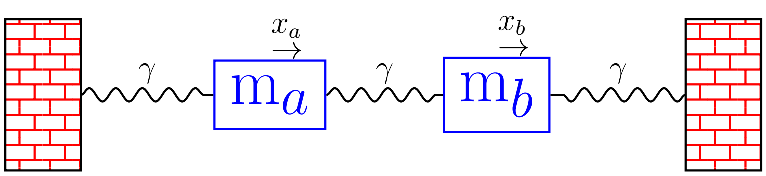
This is similar to what we did in class but we have three masses.

(a) Construct the stiffness matrix.

(b) Find the Eigenvalues.

(c) Find the Eigenvectors.

2. Consider the two mass problem where the masses are different.



Assume that $m_a = m_o$ and $m_b = \ell m_o$, where ℓ is some unitless parameter varying between 1-100.

(a) Find the eigenvalues in terms of the system constants γ, m_o, and ℓ.

(b) Plot the two eigenvalues as a function of ℓ. Can you rationalize the behavior?

(c) Assuming $\ell = 100$, $m_o = 1$, and $\gamma = 1$, plot the two eigenvectors.

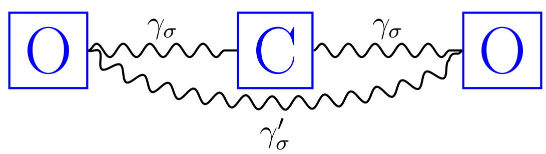
3. OPTIONAL PROBLEM: Let us consider anharmonic terms in the two mass problem from question 2 above, but let us restrict to the simpler case of $m_a = m_b$ You will soon understand enough group theory to prove that there are only two allowable interactions at third order:

$$H_{1\alpha} = \alpha(x_b - x_a)^3 \qquad\qquad H_{1\gamma} = \gamma(x_b + x_a)^2(x_b - x_a) \tag{1}$$

(a) Start by writing a molecular dynamics code for the quadratic case, assuming $m_a = m_b = 1$ and $\gamma = 1$. Generate the position and velocity trajectory, Fourier transform it as in the notes, and make sure you find peaks at $\omega = 1, \sqrt{3}$. When you choose your initial displacements, do not displace purely along one eigenmode. A harmonic system is not ergodic, so if you put all the energy in one mode, it will stay there: the other eigenmode will never be excited.

(b) Now add in the anharmonic term $H_{1\gamma}$, with $\gamma = 0.5$. Start by taking a very small displacement of just one mass. You should see approximately $\omega = 1, \sqrt{3}$ via Fourier transform of the velocity. Now slowly increase the initial displacement and monitor what happens the the frequency distribution. Plot the frequencies for several different runs to illustrate the point. If I start with $x_b = -0.12$, I am seeing some very cool behavior.

4. OPTIONAL PROBLEM using DFT: Consider CO_2 molecule, schematically pictured below.



To keep matters simple, let us only allow motion in one dimension for the time being. This will allow for the two stretching modes, but not the bending modes. Using group theory, one can show that there are 2 unique spring constants for 1d motion: $\gamma_\sigma, \gamma'_\sigma$. The corresponding potential is:

$$V = \frac{1}{2}\left[\gamma_\sigma(u_{O_1,x} - u_{C,x})^2 + \gamma_\sigma(u_{O_{\bar{1}},x} - u_{C,x})^2 + \gamma'_\sigma(u_{O_1,x} - u_{O_{\bar{1}},x})^2\right]$$

To extract all of the coefficients, you can simply compute the two cross terms:

$$\frac{\partial^2 V}{\partial u_{O_1,x}\partial u_{O_{\bar{1}},x}} \qquad\qquad \frac{\partial^2 V}{\partial u_{O_1,x}\partial u_{C_x}}$$

First relax the molecule and see how close you come to the experimental bond lenght of $1.16\mathring{A}$. Then compute the force constants. You can exploit the fact that Hellman-Feynman gives us the forces:

$$\frac{\partial^2 V}{\partial u_{O_1,x}\partial u_{C_x}} = \frac{1}{\delta}\left[\frac{\partial V(u_{O_1,x} = \delta)}{\partial u_{C_x}} - \frac{\partial V(u_{O_1,x} = 0)}{\partial u_{C_x}}\right] = \frac{1}{\delta}\left[-\vec{F}_{C_x}(u_{O_1,x} = \delta) + 0\right]$$

Therefore, you can get both force constants with a single calculation. You will need to play with δ. Try different values and make sure you are in the linear regime where the results are no longer changing very much. If you make δ too small, you will have lots of noise; if you make it too big, then you will sample cubic and quartic terms. The experimental values are $\omega_A = 1480\text{cm}^{-1}$ and $\omega_A = 2565\text{cm}^{-1}$.

- Please list the bond length you get.
- Please plot the two force constants as a function of δ to prove you see a plataue.
- Please use your force constants to compute the two frequencies, and compare to experiment.